Huawei Access Node MAR-ACC-26 Minor QoS Congestion Event SOP

Node: MAR-ACC-26 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_4869

Description

The Huawei Access Node MAR-ACC-26 at Marathahalli Core is experiencing a minor QoS congestion event, specifically a microburst-induced interface tail drop and QoS queue starvation. This failure domain is rooted in the interaction between the Access Node's Quality of Service (QoS) mechanisms and the bursty traffic patterns on the network. The Huawei Access Node, as an Access Layer device, employs a combination of hardware and software-based QoS solutions to manage traffic congestion. However, the presence of microbursts can overwhelm the node's buffer capacity, causing interface tail drops and QoS queue starvation. Furthermore, the Huawei Access Node's QoS architecture relies on a complex interplay between traffic policing, shaping, and scheduling mechanisms to ensure optimal traffic handling. In the event of a QoS congestion event, the node's QoS policies may not be effective in mitigating the effects of microbursts, leading to packet loss and QoS queue starvation. To resolve this issue, it is essential to identify the root cause of the QoS congestion event and implement corrective actions to address the underlying issues.

Pre-requisite Checks

- Confirm the Node ID, Site Location, and Vendor (CLI Dialect) for the affected Access Node.
- Verify the Role and Failure Domain to ensure the correct scope of work.
- Check the redundant link status to ensure that the issue is not due to a single point of failure.
- Dispatch remote hands to Marathahalli Core to facilitate diagnostic and resolution efforts.

Potential Root Causes

Cause	Probability
Microburst-induced congestion on the Access Node's interface	High
Insufficient buffer capacity on the Access Node's interface	Medium
Ineffective QoS policies on the Access Node	Low

Diagnostic Commands & Expected Outputs

```
Command: display interface GigabitEthernet 1/0/1, Expected Output: /GigabitEthernet 1/0/1 is up, line-protocol is up\n Current address: 192.168.1.1\n Last change: 2023-02-20 14:30:00, Description: This command displays the interface status and configuration for GigabitEthernet 1/0/1, which may help identify any issues with the interface.
```

```
Command: display qos profile, Expected Output: /QoS profile:\n Profile name: DEFAULT\n Traffic policing: enabled\n Traffic shaping: disabled, Description: This command displays the QoS profile configuration, which may help identify any issues with QoS policy enforcement.

Command: display queueing GigabitEthernet 1/0/1, Expected Output: /GigabitEthernet 1/0/1 queueing:\n Queue type: PQ\n Queue size: 1000, Description: This command displays the queueing configuration for GigabitEthernet 1/0/1, which may help identify any issues with queue management.
```

Resolution Steps

Step 1: Step: Verify the interface status and configuration, Command: display interface GigabitEthernet 1/0/1, Description: Verify that the interface is up and that the configuration is correct.

Step 2: Step: Verify the QoS profile configuration, Command: display qos profile, Description: Verify that the QoS profile is configured correctly and that traffic policing is enabled.

Step 3: Step: Adjust the queue size, Command: queueing GigabitEthernet 1/0/1 queue-size 2000, Description: Adjust the queue size to a larger value to provide more buffer capacity for the interface.

Post-Check Verification

```
Command: display interface GigabitEthernet 1/0/1, Expected Output: /GigabitEthernet 1/0/1 is up, line-protocol is up\n Current address: 192.168.1.1\n Last change: 2023-02-20 14:30:00, Description: Verify that the interface status and configuration are correct after the resolution steps.

Command: display qos profile, Expected Output: /QoS profile:\n Profile name: DEFAULT\n Traffic policing: enabled\n Traffic shaping: disabled, Description: Verify that the QoS profile configuration is correct after the resolution steps.

Command: display queueing GigabitEthernet 1/0/1, Expected Output: /GigabitEthernet 1/0/1 queueing:\n Queue type: PQ\n Queue size: 2000, Description: Verify that the queue size has been adjusted correctly after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original queue size, Command: queueing GigabitEthernet 1/0/1 queue-size 1000, Description: Restore the original queue size if the resolution steps fail.